

Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Integration

| HTTP Request between ERPNext and Node-RED

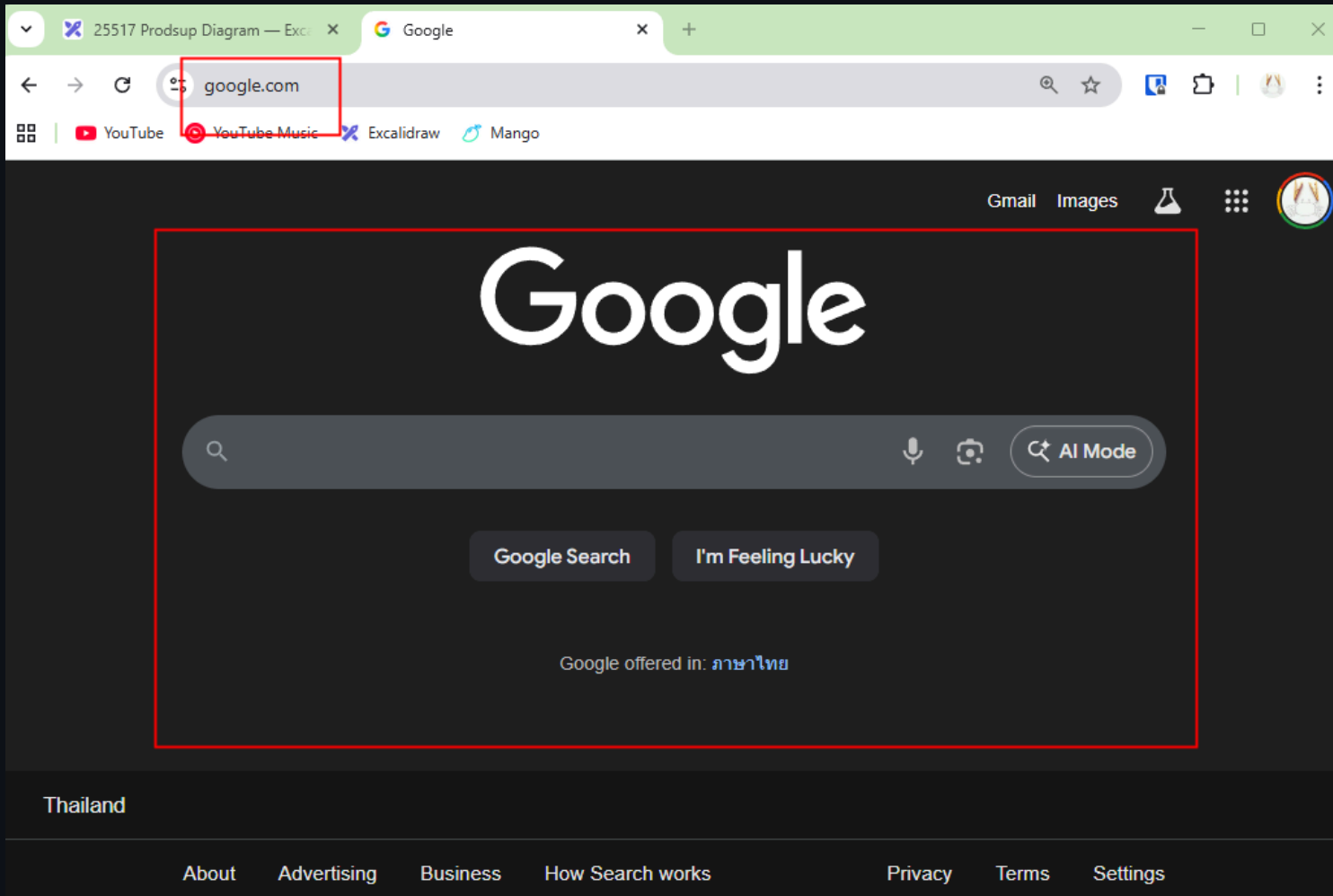
What is HTTP Request?

- HTTP (Hypertext Transfer Protocol) is the foundation of data communication on the World Wide Web.
- We can connect ERPNext and Node-RED via HTTP Request to exchange data.

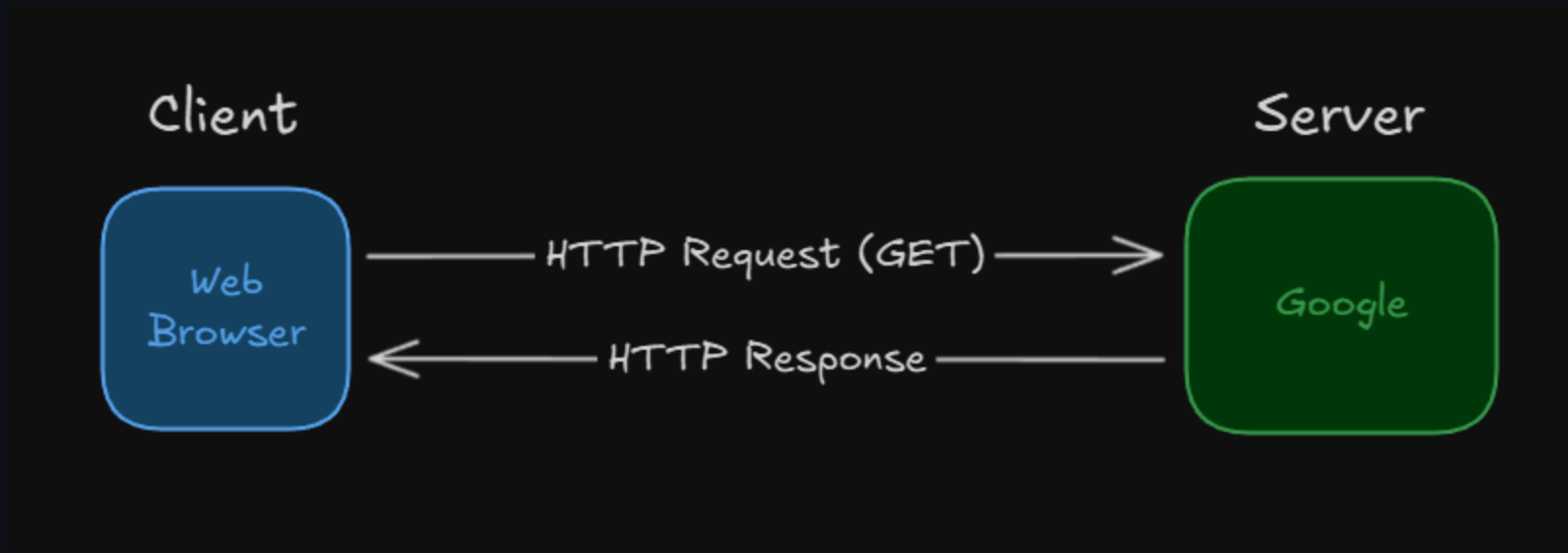
HTTP Methods

- GET : Retrieve data from a server.
- POST : Send data to a server to create a resource.

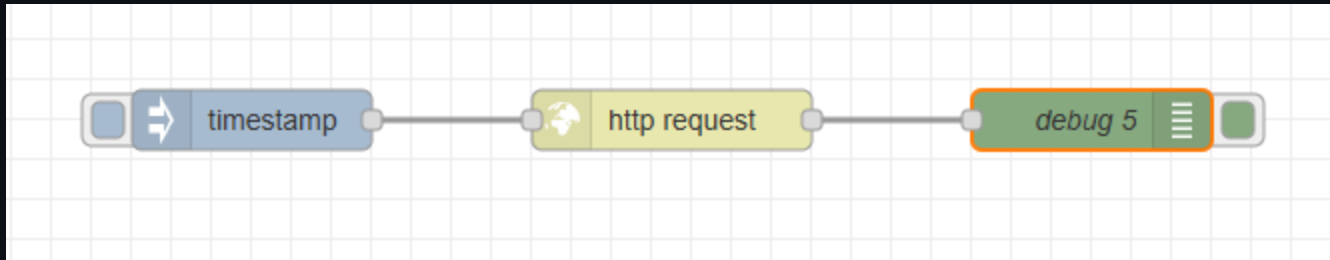
HTTP GET Request Example



Role

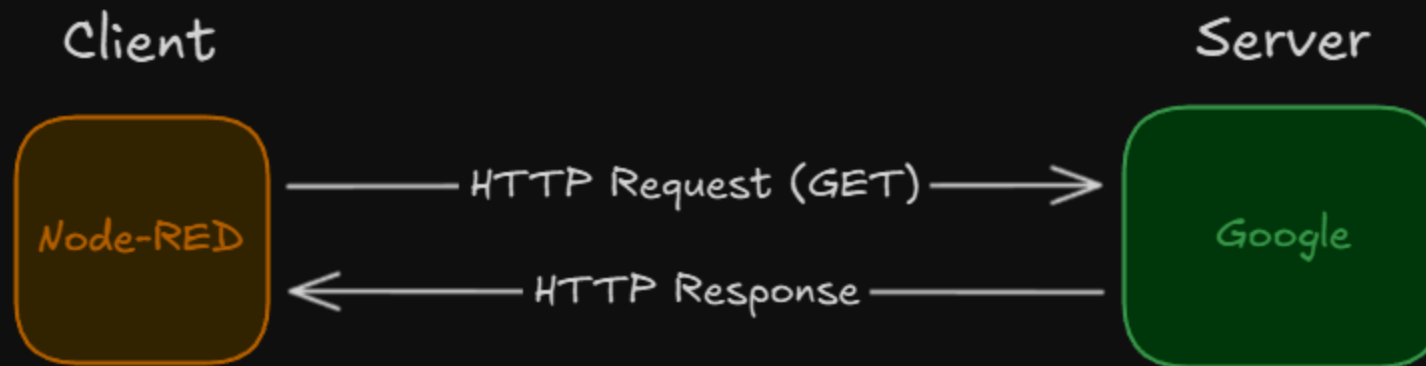


Part 1: Node-RED to Google

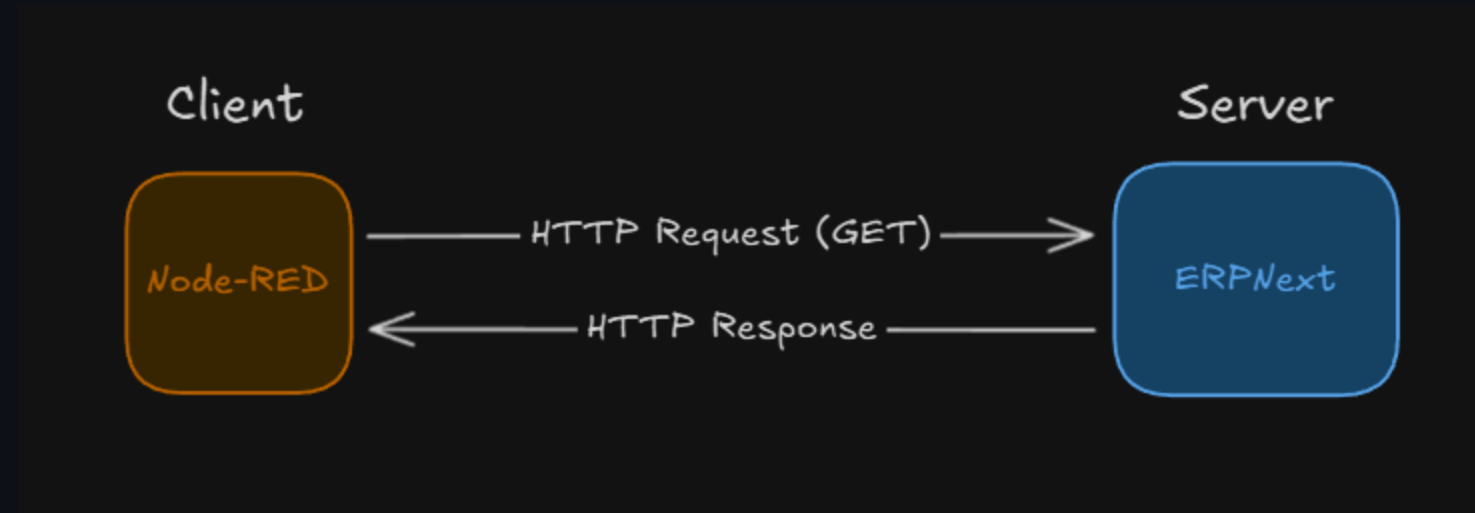


- Method: GET
- URL: <https://www.google.com>

Role



Part 2: Node-RED to ERPNext



- We have to use [ERPNext API](#).
 - *Application Programming Interface*

ERPNext API Authentication

- ERPNext API requires authentication via `API Key` and `API Secret`.
- These can be generated from the User Settings in ERPNext.
- The `API Key` and `API Secret` are then combined to form a token for authorization.
- This token is included in the HTTP request **headers** to authenticate the request.

Get API Key and Secret

The screenshot shows the Frappe Framework user interface. On the left sidebar, the 'Frappe Framework' logo is at the top, followed by navigation links: 'Build', 'Tools', 'Users' (highlighted with a red box), 'Website', and 'Integrations'. The main header shows the breadcrumb 'Users / User / erp_admin@cmu.com'. Below this, the user's name 'Administrator' is displayed with an 'Active' status tag. A red box highlights the 'Administrator' text. The 'Settings' tab is selected and highlighted with a red box. Below the tabs, there are sections for 'App', 'Security Settings', and 'Third Party Authentication'. The 'API Access' section is expanded, showing the 'API Key' as '477256912f164d4' and the 'API Secret' as a masked string '*****'. A red box highlights the entire 'API Access' section. Below the API key, it states 'API Key cannot be regenerated'. At the bottom of the API Access section, there is a 'Generate Keys' button and a link 'Click here to learn about token-based authentication'.

Frappe Framework

Users / User / erp_admin@cmu.com

Administrator Active

User Details Roles & Permissions More Information **Settings** Connections Sessions

App ▾

Security Settings ▾

Third Party Authentication ▾

API Access ^

API Key API Secret

477256912f164d4 *****

API Key cannot be regenerated

Generate Keys

[Click here to learn about token-based authentication](#)

Construct Token

```
KEY=  
SECRET=  
TOKEN=token ${KEY}:${SECRET}
```

Note: The word "token" is required. There is a space between `token` and `${KEY}`.

Construct URL

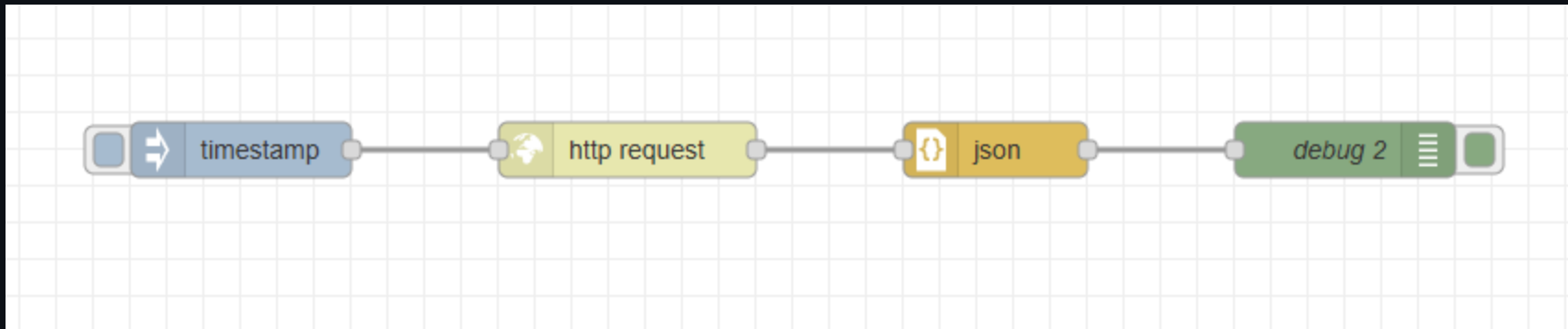
```
BASE_URL=https://pmX-erpXX.iecmu.com
```

```
URL1=${BASE_URL}/api/resource/Sales Order
```

```
URL2=${BASE_URL}/api/resource/Sales Order?fields=["*"]
```

```
URL3=${BASE_URL}/api/resource/Sales Order?fields=["*"]&filters=[["docstatus","=","1"]]
```

HTTP Request Flow



- Method: `GET`
- URL: `URL1` , `URL2` , or `URL3`
- Headers: `Authorization: ${TOKEN}`

Headers

☐ Enable connection keep-alive

☐ Use proxy

☐ Only send non-2xx responses to Catch node

☐ Disable strict HTTP parsing

← Return a UTF-8 string ▼

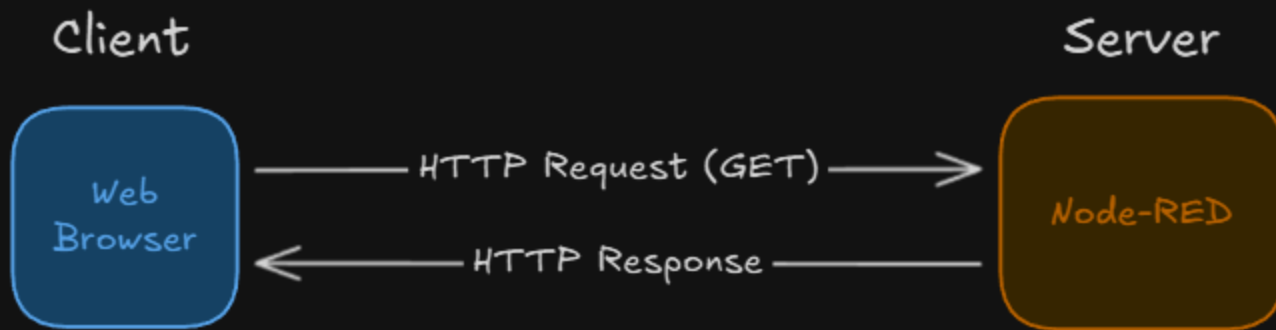
☰ Headers

☰ ▼ a_z Authorization ▼ a_z token 477256912f164d4:0f ✕

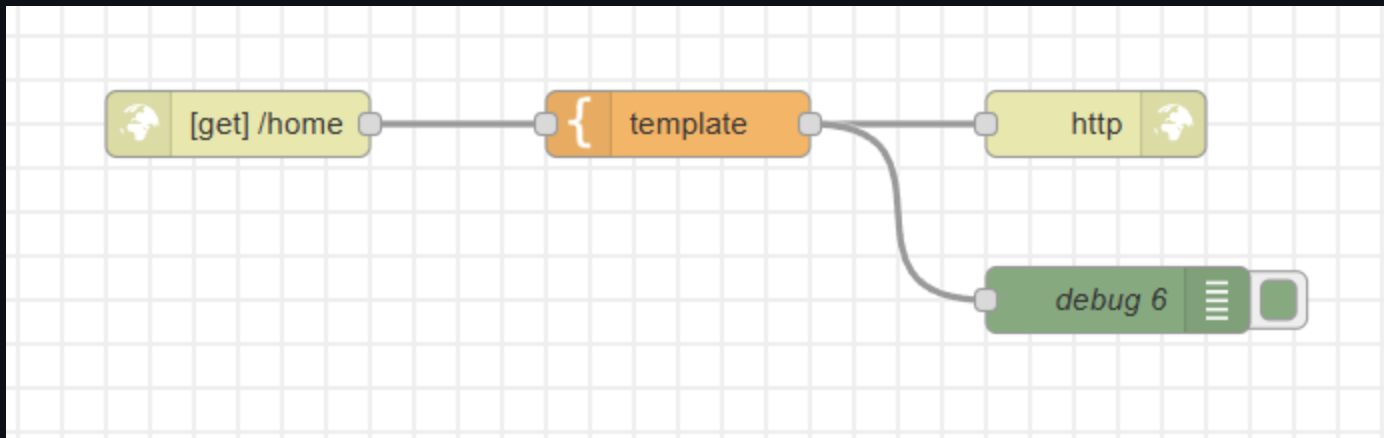
JSON?

- *JavaScript Object Notation*
- A lightweight data-interchange format that is easy for humans to read and write, and easy for machines to parse and generate.

Part 3: Node-RED as HTTP Server



Flow



HTTP In Node

Edit http in node

Delete Cancel Done

Properties

Method GET

URL home

Name Name

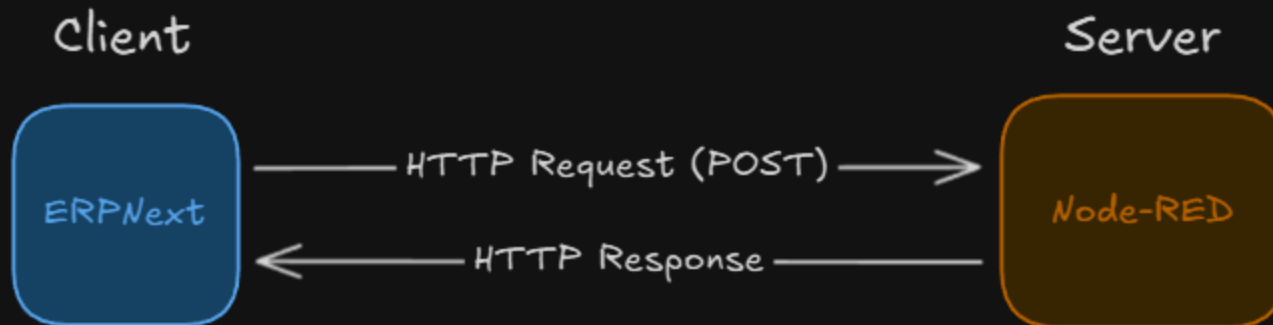
Testing

- Construct `NODE_RED_URL`

```
NODE_RED_URL=https://pmX-nrXX.iecmu.com
```

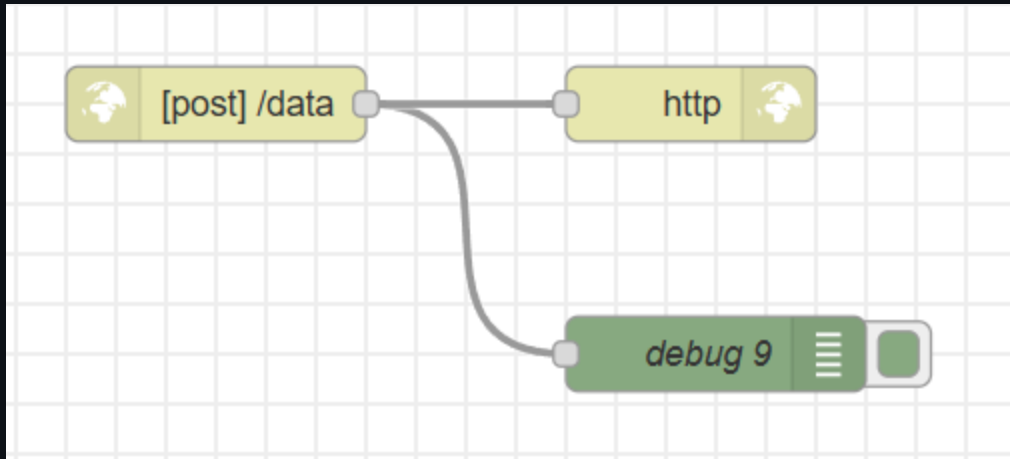
- Visit `${NODE_RED_URL}/home` in your web browser.
 - You should see the message written in `Template` Node.

Part 4: ERPNext to Node-RED



- Notice the `POST` method.

Setup Node-RED HTTP Server



HTTP-in Node

Edit http in node

Delete Cancel Done

⚙ Properties

Method POST

☐ Do not parse request body

☐ Accept file uploads

URL data

Name Name

ERPNext Webhook

- **Webhook** is a way for an application to provide other applications with real-time information.
- It delivers data to other applications as it happens, meaning you get data immediately.
- ERPNext supports **Webhook** to send data to other systems when certain events occur.

Create Webhook in ERPNext

- Go to Frappe Framework > Integration > Webhook
- Create a new Webhook with the following details:
 - Document Type: Sales Order
 - Doc Event: on_update
 - Request URL: \${NODE_RED_URL}/data
 - Request Method: POST
 - Request Structure: JSON
 - Webhook Data: {{ doc|tojson|safe }}

Test Webhook

- Create and Update a `Sales Order` document in ERPNext.
- Check the Node-RED Debug panel to see if the data is received.
- Note
 - Once you submit the Sales Order, further updates will not trigger the webhook.
 - `on_update` event is only triggered when the document is modified in the draft state.